

Drought-related cumulative and time-lag effects on vegetation dynamics across the Yellow River Basin, China

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ARTICLE INFO

Keywords:

Drought
NDVI
Cumulative effect
Time-lag effect
Water availability
Yellow River Basin

ABSTRACT

Yellow River Basin (YRB), a climate-sensitive and ecologically compromised area in China, is increasingly affected by extreme climate events (especially droughts) resulting from climate change and frequent human activity. Vegetation responds asymmetrically to drought with cumulative and time-lag effects, whereas response across various climatic zones and diverse vegetation types in the YRB remains unclear. To address this deficiency, we examined the spatiotemporal patterns of accumulated and lagged drought effects on vegetation dynamics for the period 1982 to 2015. The examination was based on the long-term Normalized Difference Vegetation Index (NDVI) and multiscale dataset of the Standardized Precipitation Evapotranspiration Index (SPEI). Cumulative (time-lag) effects were determined via the maximum correlation between the NDVI and the one- to 12-month timescale SPEI (one-month timescale SPEI), as well as the corresponding months of optimal response to drought. The main findings were as follows: (1) Accumulated and lagged drought significantly affected approximately 50% and 60% of the vegetated area in the YRB, respectively, with the strongest effects and the corresponding optimal months varying across climatic zones and vegetation types. (2) In general, vegetation in the arid zone tended to be more sensitive and resistant to drought, as evidenced by the occurrence of lagged and accumulated drought effects mostly in the short-term (one–three months) and medium-term (six–eight months), respectively. This finding may be related to the vegetation's strategy for coping with water deficits. (3) The biome-level effects of drought on grassland and cultivated vegetation were stronger than those on forests, which may be associated with differences in the functional characteristics of root systems. (4) Annual water availability significantly affected the spatiotemporal patterns by which the NDVI responded to droughts of multiple time-scales, with the correlation coefficients and corresponding months of response decreasing with increasing average annual SPEI. These results indicate that vegetated areas with low water availability were more susceptible to cumulative (time-lag) droughts. (5) Independent of the climate zone or vegetation type, drought cumulatively affected vegetation more than time-lag effects. The study improves the knowledge of climate–vegetation relationships in the YRB and provides theoretical support for addressing drought risk in a changing climate.

1. Introduction

As the principal component of terrestrial ecosystems, vegetation is vital for regulating climate and maintaining the water–carbon balance (Liu et al., 2021; Xu et al., 2019). It is commonly applied as an indicator of variations in ecosystems, because vegetation dynamics are vulnerable to climate change, particularly extreme climate events (Luo et al., 2020; Ying et al., 2020). Recent decades have witnessed an increase in the frequency and intensity of extreme climate, such as droughts, floods, and cold waves (AghaKouchak et al., 2020; Bellprat et al., 2019;

Ummenhofer and Meehl, 2017). Of these phenomena, the most common is drought, which is a condition of prolonged and significantly lower water availability than the normal level, thereby causing a failure to supply water demand (Vicente-Serrano et al., 2013). Water availability is intimately related to vegetation growth, and chronically low water supplies may adversely affect vegetation function, even resulting in vegetation death (Gao et al., 2021; Zhao et al., 2021). Additionally, drought substantially affects vegetation at various spatial and temporal scales because of its high frequency, long duration, widespread effects, and multiple timescales (Wang et al., 2019; Zhang et al., 2017). The

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<https://doi.org/10.1016/j.ecolind.2022.109409>

Received 2 August 2022; Received in revised form 30 August 2022; Accepted 31 August 2022

Available online 7 September 2022

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above issues point to the need to investigate the association between vegetation dynamics and drought to facilitate the understanding of vegetation sensitivity to and drought tolerance (Anderegg et al., 2020; Fang and Xiong, 2015).

Given that the spatiotemporal patterns of vegetation–drought interactions differ significantly depending on vegetation type, climate condition, and timescale, gaining a comprehensive grasp of how drought affects vegetation is difficult (Li et al., 2018; Zhang et al., 2017). Among these variations, those in the timescales of drought that affect vegetation are one of the most significant mechanisms of climate–vegetation interactions (Jiang et al., 2020; Qi et al., 2022). Although several studies have focused on the response of vegetation to simultaneous or single-timescale droughts (Jiang et al., 2018; Mohammad et al., 2013; Wang et al., 2021; Zhao et al., 2019), none of these explorations considered the extensive acknowledgment of vegetation as affected by the temporal effects of drought, including cumulative and time-lag effects (Hua et al., 2019; Zhang et al., 2022; Zuo et al., 2021). Accordingly, identifying the temporal variability of vegetation responses to drought is crucial to expanding the knowledge of the mechanisms that underlie climate–vegetation interactions and developing effective vegetation conservation strategies.

Time-lag effects describe how previous droughts affect current vegetation at a specific timescale, thus reflecting the drought sensitivity of vegetation (D'Orangeville et al., 2018; Wei et al., 2022a). Studies have extensively investigated the effects of lagged drought on vegetation growth, including Peng et al. (2019), who found that 46.2 % of the vegetated areas over the extratropical ecosystems in the Northern Hemisphere reflect the impact of lagged drought on autumn leaf senescence. The authors further discovered that in which the lagged months periods occurred mainly within six months. In Nebraska, Hua et al. (2019) found that vegetation typically responded in a lagged manner to drought within 30 to 45 days and that this response differed depending on vegetation type. Zhao et al. (2020) reported that nearly half of the grassland on the Loess Plateau continued to exhibit severe effects from a drought event that had occurred three months before. A similar legacy effect was observed by Xu et al. (2021), who indicated that 25 % of the vegetated area in northern China exhibited a predominant lag in effects of 9.1 months.

Studies have also identified that a long-term deficiency in atmospheric and soil water availability (cumulative drought) significantly affects vegetation growth, inhibiting photosynthesis, reducing vegetation productivity, and even causing vegetation death (Wei et al., 2022a; Xu et al., 2021; Zhang et al., 2022). Cumulative drought refers to the effects of a period (spanning several months or even years) of water deficits on vegetation growth and can be used to assess vegetation tolerance to drought (Zhang et al., 2022). Peng et al. (2019) uncovered cumulative drought effects on 27.2 % of vegetated areas over the Northern Hemisphere at the end of the growing season, with the corresponding period of accumulated effects being one to four months. Zhao et al. (2020) reported that approximately 81 % of grasslands on the Loess Plateau are affected by cumulative drought, with accumulated effects occurring primarily from May to October. Wei et al. (2022a) revealed that 78.5 % of global grasslands are affected by cumulative drought and that the period of accumulated effects is longer in arid zones (eight–10 months) than in humid regions (less than six months). As indicated by Zhang et al. (2022), cumulative droughts have a wide range of effects on global vegetation, with a typical cumulative effect duration (CED) of 4.89 months. The authors also stated that CED is spatially and temporally heterogeneous across vegetation types, water balance levels, and elevation levels, among other factors.

As a climate-sensitive area, the Yellow River Basin (YRB) in China covers arid, semi-arid, and semi-humid zones, with most areas receiving <500 mm of annual precipitation (Zhan et al., 2022). The YRB is one of the most severely eroded areas in the world, with vegetation in the basin vulnerable to multiscale droughts (Qin et al., 2022). Several studies have assessed the correlation between vegetation and simultaneous drought

in the YRB, leading to a bias in understanding vegetation sensitivity and tolerance to multiscale drought (Geng et al., 2022; Jia et al., 2021; Wang et al., 2015; Zhao et al., 2018). Some studies have evaluated the temporal effects of drought on vegetation on the basis of either a single species or an entire community of vegetation (Jiang et al., 2022; Wei et al., 2022a; Zhao et al., 2020), which ignored the variability of vegetation responses to drought across climate zones and vegetation types.

To bridge the above-mentioned gap, we used the long-term Normalized Difference Vegetation Index (NDVI) and the lagged and accumulated Standardized Precipitation Evapotranspiration Index (SPEI) series of the YRB to (1) determine spatiotemporal patterns typifying the cumulative and time-lag effects of drought on vegetation dynamics for various vegetation types and climatic conditions and (2) identify the relationship between the intensity and duration of accumulated (lagged) drought effects on vegetation and annual water availability.

2. Materials and methods

2.1. Study area

The YRB originates in the Bayan Har Mountains of the Qinghai–Tibet Plateau and flows through nine provinces (autonomous regions) in China, covering an area of $79.5 \times 10^4 \text{ km}^2$ (Ji et al., 2018). Its topography decreases from east to west and sequentially spans four geomorphological units: the Tibetan Plateau, Inner Mongolia Plateau, Loess Plateau, and Huang-Huai-Hai Plain. According to the ecogeographical classification of China developed by Zheng (1999), the basin can be divided into three climatic zones, being arid in the northwest, semi-arid in the center, and semi-humid in the southeast. Approximately 75 % of the vegetated area in the YRB consists of cultivated lands, grasslands, and meadows. Cultivated lands are predominantly located in the southeastern semi-arid zone and semi-humid zone, grasslands are found in the arid zone and northeastern semi-arid zone, and meadows are situated on the Qinghai–Tibet Plateau (Fig. 1).

2.2. Data

2.2.1. NDVI dataset

The NDVI3g.v1 dataset released by the Global Inventory Monitoring and Modeling System (GIMMS, <https://ecocast.arc.nasa.gov/data/pub/gimms/3g.v1/>), was utilized in this study. The dataset has a spatial resolution of $8 \text{ km} \times 8 \text{ km}$ with a 15-day interval and covers the period 1982 to 2015. To eliminate the effects of clouds, aerosols, and other non-vegetation factors on remote sensing for monitoring, we calculated the monthly NDVI by the maximum value composite approach. To mitigate the effects of soil background on the NDVI in areas with low vegetation cover, pixels with multi-year average NDVI values <0.1 were considered unvegetated regions (Jin et al., 2022).

2.2.2. Drought index

We obtained the monthly SPEI with a spatial resolution of 0.5° at one- to 12-month timescales during 1981–2015 from the SPEIbase (version 2.6) of Consejo Superior de Investigaciones Científicas (https://spei.csic.es/spei_database/). The SPEI reflects the moisture surplus or deficit divided by the monthly moisture deficit [precipitation minus potential evapotranspiration (PET)], where PET is calculated using the FAO-56 Penman–Monteith equation (Allen et al., 1998; Vicente-Serrano et al., 2010). Assuming that the cumulative moisture deficit in a single month of historical data follows a log-logistic distribution, the probability density function and cumulative probability are calculated and transformed into a standard normal distribution to obtain a multiscale SPEI. The one-month and 12-month SPEI timescales (e.g., in December), for example, are indicators of the atmospheric water balance during the current month and entire year (accumulated from one to 12 months), respectively (Vicente-Serrano et al., 2013). In this paper, we

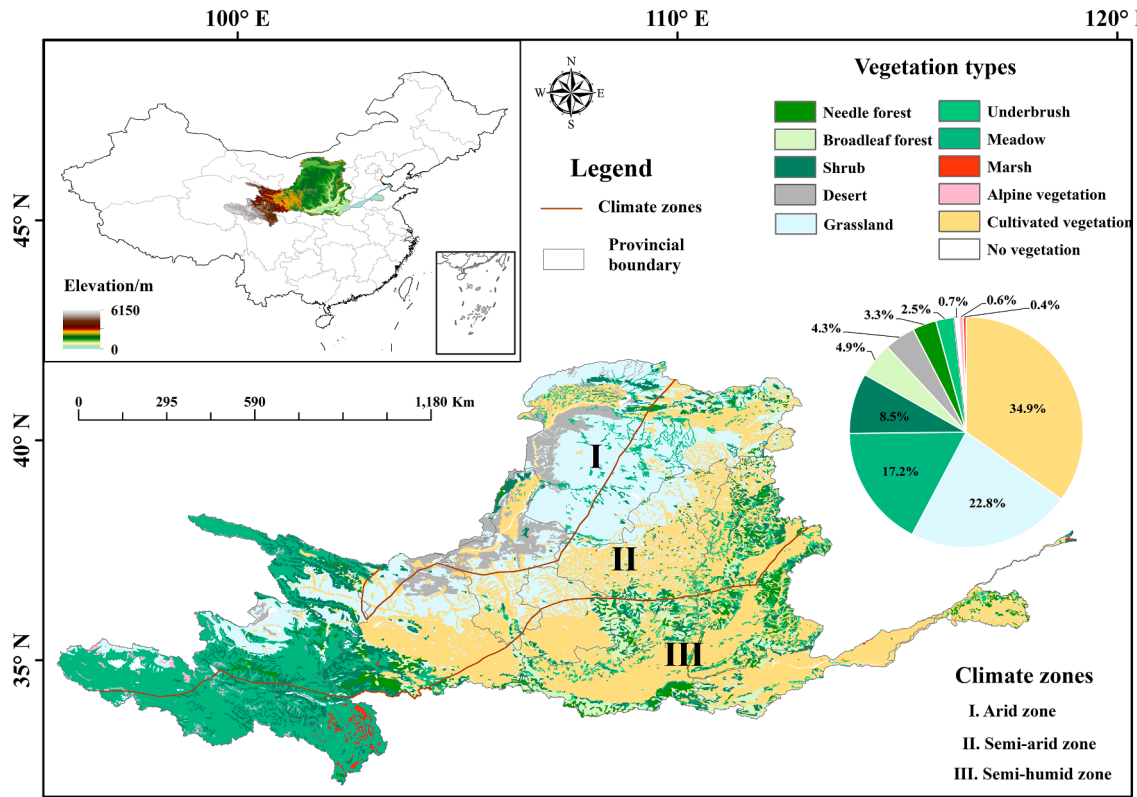


Fig. 1. Map of climate zones and vegetation types in the YRB.

investigated monthly SPEI for the 1- to 12- timescale from 1981 to 2015 to quantify how the accumulated (lagged) drought affect vegetation in the YRB.

2.2.3. Vegetation types and climate zones

A spatial distribution dataset of vegetation types (1:1,000,000 scale) in China was provided by the Resource and Environment Science and

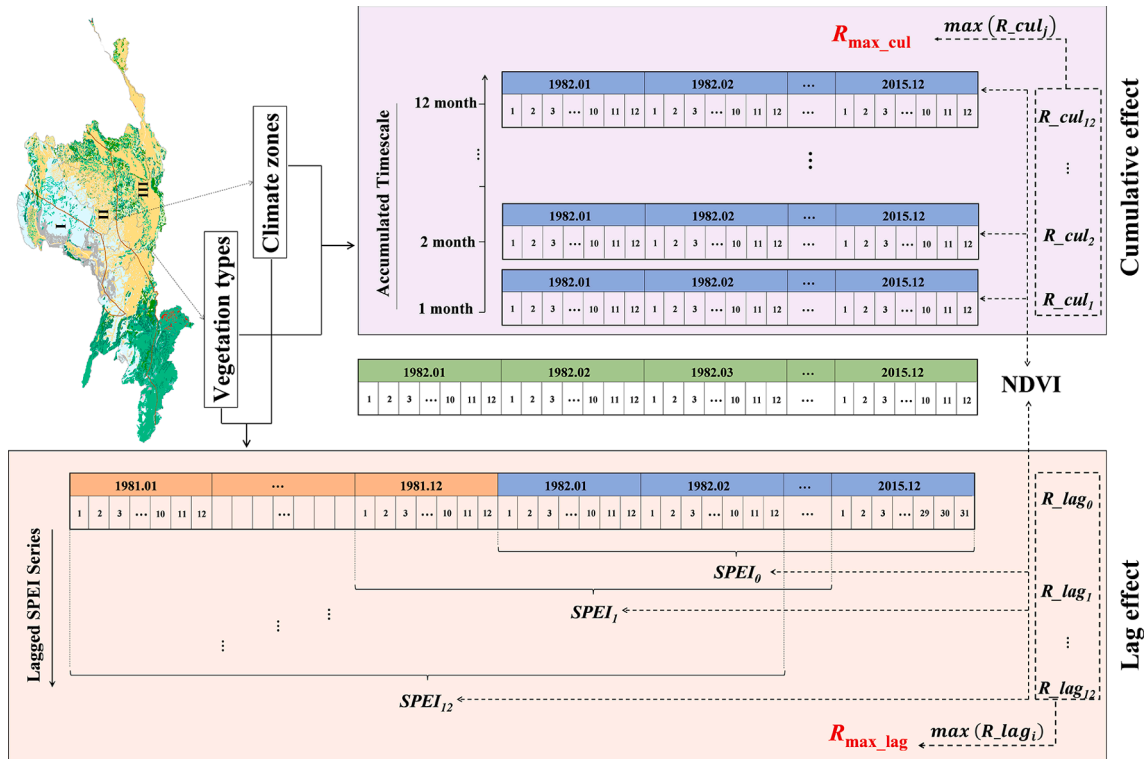


Fig. 2. Schematic illustrating the determination of the strongest lagged (cumulative) effects and the corresponding months of optimal responses to lagged (accumulated) effects in the YRB.

Data Center (RESDC, <https://www.resdc.cn>) (Xueyu, 2019). This dataset is spatially resolved at a resolution of 1 km and comprises data on vegetation units of 11 vegetation types: needle forest, broadleaf forest, shrub, desert, grassland, underbrush, meadow, marsh, alpine vegetation, cultivated vegetation, and others (Fig. 1).

The climatic zones were derived from the “Ecogeographic Zoning Data for China” of the RESDC, which classified China into four climatic zones based on its wet and dry conditions, including humid, semi-humid, semi-arid, and arid zones. According to this criterion, the YRB was classified as arid, semi-arid, and semi-humid.

2.3. Methods

2.3.1. Cumulative effects of drought on vegetation

As illustrated in Fig. 2, We used the Pearson correlation coefficient (R), we can determine whether a negative ($R < 0$) or positive ($R > 0$) relationship exists between drought and vegetation growth, with a higher value of R indicating a more significant impact (Zhao et al., 2020). We calculated all the significant correlation coefficients (R_{culj} , statistical significance was determined at $p < 0.05$) between the accumulated drought series ($SPEI_j$, $1 \leq j \leq 12$) and monthly NDVI as candidate indicators in the effects of accumulated drought on vegetation [Equation (1)], after which we identified the months of optimal response to accumulated drought effects and the corresponding intensity of the time accumulation for each pixel in accordance with the maximum R_{culj} (i.e., R_{max_cul} at $p < 0.05$) [Equation (2)].

$$R_{culj} = \text{corr}(NDVI, SPEI_j) \quad 1 \leq j \leq 12 \quad (1)$$

$$R_{max_cul} = \max(R_{culj}) \quad 1 \leq j \leq 12 \quad (2)$$

In Equations (1) and (2), R_{culj} is the Pearson coefficient of correlation between the NDVI and accumulated SPEI, j is the timescale of accumulated effects from one- to 12- months, and R_{max_cul} denotes the maximum absolute value of R_{culj} .

2.3.2. Lagged effects of drought on vegetation

In contrast to the examination of the cumulative effects, for which one- to 12-month timescale SPEI was used, the lag effects involved a consideration of only the one-month timescale SPEI. We calculated correlation coefficients (R_{lagi} , statistical significance was determined at $p < 0.05$) with different lagged SPEI series [Equation (3)], and the lagged effect was determined at this pixel by considering the correlation coefficient between NDVI and accumulated SPEI that was significant. Then we determined the optimal lagged months and the corresponding intensity of the lagged effects for each pixel with the maximum R_{lagi} (i.e., R_{max_lag} at $p < 0.05$) [Equation (4)].

$$R_{lagi} = \text{corr}(NDVI, SPEI_i) \quad 0 \leq i \leq 12 \quad (3)$$

$$R_{max_lag} = \max(R_{lagi}) \quad 0 \leq i \leq 12 \quad (4)$$

In the equations above, R_{lagi} is the Pearson coefficient of correlation between the NDVI and lagged SPEI series, $SPEI_i$ denotes the one-month SPEI timescale at lagged i month. The values of i ranges from 0 to 12, where $i = 0$ indicates no time-lag effect, while others indicate a time-lag of one- to 12- month. For example, a lag of one month was primarily analysed using the monthly SPEI from December 1981 to November 2015 correlated with the monthly NDVI over 1982–2015, and similarly to a lag of other lagged months (Fig. 2). R_{max_lag} represents the maximum value of R_{lagi} , with the month in which R_{max_lag} was recorded was considered the optimal lagged month.

2.3.3. Influence of hydrological conditions on cumulative and lagged effects

Since SPEI was calculated based on the difference between monthly precipitation and PET, it is capable of reflecting the water balance and is primarily responsible for the observable differences in vegetation

resistance to drought events and post-drought recovery capacities between dry and wet conditions (Vicente-Serrano et al., 2013). Furthermore, the 12-month timescale SPEI in December was indicative of the annual water availability (water balance), which could be used to quantify the impact of hydrological conditions on these cumulative and time-lag effects (Peng et al., 2019; Wei et al., 2022a; Zhao et al., 2020). Therefore, to identify the underlying influence of hydrological conditions on temporal effects, we evaluated the correlation between mean values of R_{culj} (R_{lagi}) and corresponding accumulated (lagged) months to annual SPEI in various climate zones and vegetation types within the YRB.

3. Results

3.1. Cumulative effects of drought on vegetation dynamics

3.1.1. Temporal variation of cumulative effects

We quantified the cumulative effects of drought on the NDVI in the YRB based on the R_{culj} between the NDVI and the one- to 12-month SPEI for the period 1982 to 2015. The months of accumulated effects corresponded to the maximum R_{culj} (R_{max_cul}), indicating drought tolerance, with longer accumulation reflecting greater tolerance. Drought tolerance in the YRB varied considerably across vegetation types and climatic zones (Fig. 3). The mean R_{max_cul} and vegetated areas significantly affected by the accumulated SPEI in the arid zone increased with months of accumulated effects, peaking at seven months before gradually decreasing [Fig. 3(a)]. A similar temporal variation in R_{max_cul} was recorded in the semi-arid zone in comparison to the arid area, but the peak was observed in the accumulation period of five months. In contrast, two peaks of R_{max_cul} occurred in the semi-humid zone, specifically at months 2 and 8 of accumulated effects, respectively. In general, the R_{max_cul} values and significantly correlated areas were ranked in the order arid zone > semi-humid zone > semi-arid zone, independent of month of accumulated effects.

Furthermore, the vegetated areas significantly affected by accumulated drought fluctuated across vegetation types, illustrating variability in drought resilience [Fig. 3(b)]. Alpine vegetation and meadows were the least tolerant to drought, as these vegetated areas were affected by simultaneous SPEI, and the corresponding R_{max_cul} values were greater than those affected by cumulative drought. On the whole, grasslands and cultivated vegetation responded most strongly to short- to medium-term cumulative drought (accumulated effects over four–six months), whereas shrubs, conifers, and broadleaf forests were most severely affected by medium- to long-term cumulative drought (accumulated effects over seven–nine months).

3.1.2. Spatial patterns of the cumulative effect

The spatial patterns of the R_{max_cul} were shown in Fig. 4, along with the corresponding months of accumulated effects. The NDVI and accumulated SPEI were positively correlated in 77.4 % of the YRB, but they were negatively correlated primarily at high altitudes in the semi-humid zone [Fig. 4(a)]. Overall, nearly half of the vegetated areas in the YRB were significantly affected by cumulative drought ($p < 0.05$). The R_{max_cul} was greater than 0 in 40.4 % of these areas, predominantly in the central and western areas of the arid and semi-arid zones. Areas with $<0R_{max_cul}$ were concentrated in high-altitude regions of the semi-humid zone, with occurrence amounting to 9.4 %.

The five-month accumulated drought most extensively affected vegetation in the YRB (with the areas affected spanning 33.8 %), and the effect was distributed primarily in the northeastern part of the basin. The next strongest effects were those from four months (14.5 %) and eight months (11.7 %) of accumulated drought. Additionally, short-term accumulated drought occurred principally at high elevations in the upper basin, whereas long-term accumulated drought occurred in the southeastern semi-humid zone.

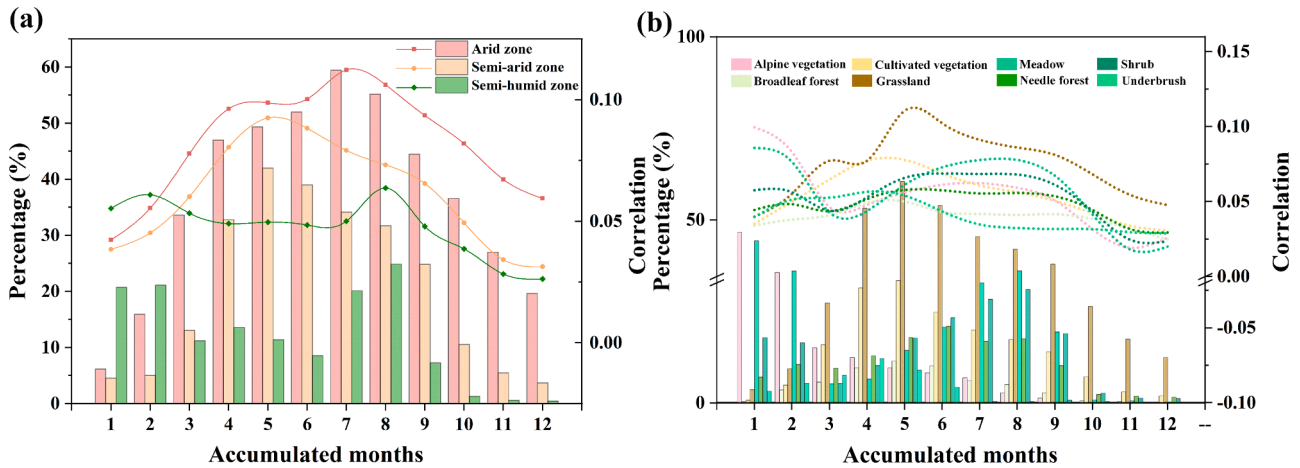


Fig. 3. Temporal variations in the (a) mean absolute values of the $R_{\max_cul}$ and (b) the areas with significant $R_{\max_cul}$.

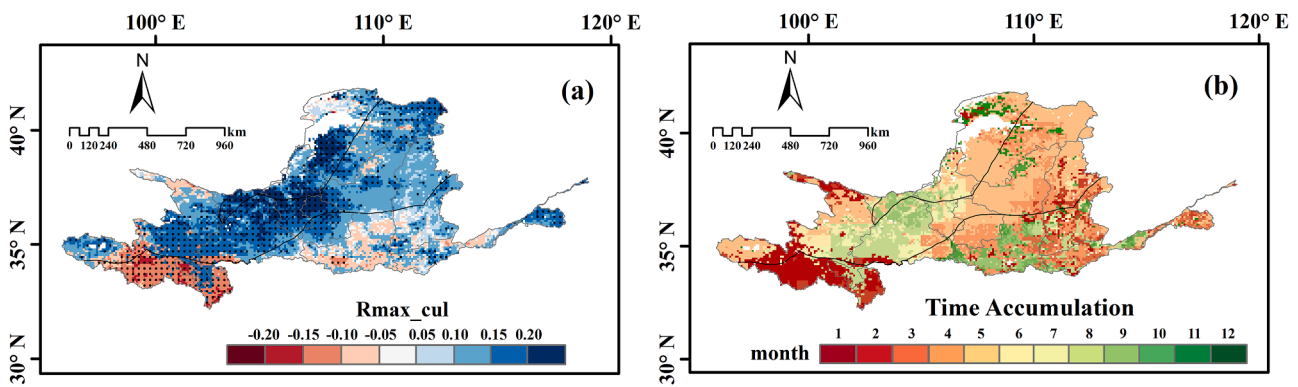


Fig. 4. Spatial patterns of (a) the $R_{\max_cul}$ and (b) corresponding months of accumulated drought in relation to the $R_{\max_cul}$.

3.1.3. Relationship between water availability and cumulative effects

The strongest response of vegetation to cumulative drought ($R_{\max_cul}$) and the corresponding mean months of accumulated effects were closely related to annual water availability in the YRB (Fig. 5). The mean $R_{\max_cul}$ was significantly negatively correlated with annual water availability ($R^2 = 0.48, p < 0.001$), indicating that vegetation in arid

zones was most strongly affected by accumulated drought [Fig. 4(a)]. A similar relationship was found with respect to the significant negative correlation between months of accumulated drought and annual water availability ($R^2 = 0.54, p < 0.001$), reflecting that the NDVI in arid zones was affected by accumulated drought over a long period [Fig. 4(b)].

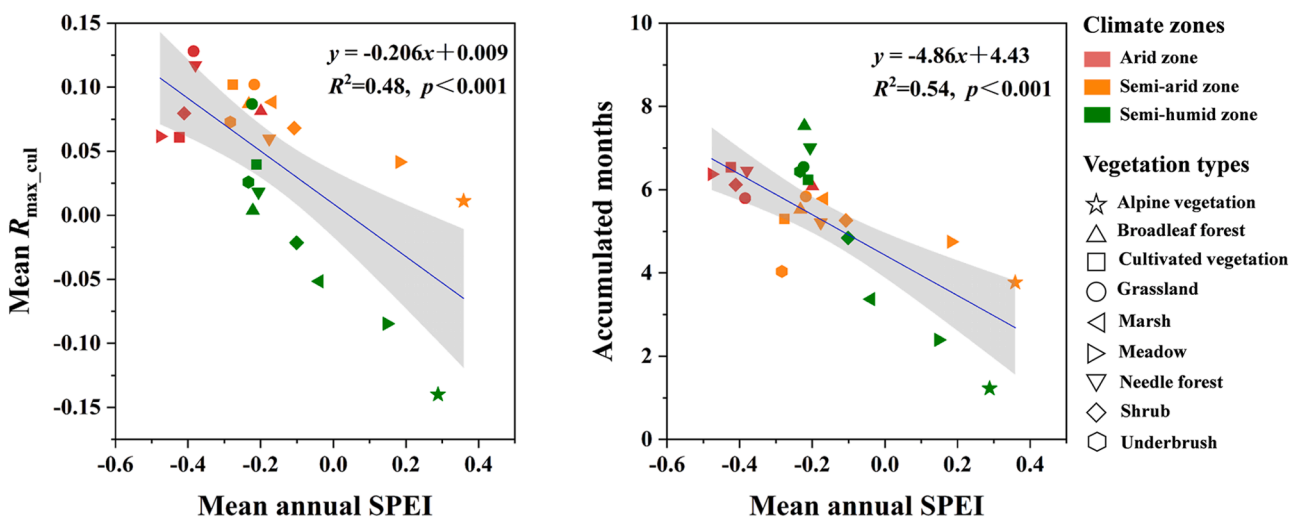


Fig. 5. The relationship between the 12-month timescale SPEI in December and (a) the mean $R_{\max_cul}$ and (b) the corresponding mean months of accumulated drought effects for different vegetation types and climatic zones. The symbols represent different vegetation types, the colors indicate different climatic zones, and the gray shaded areas refer to the 95% confidence intervals of the fitted lines.

3.2. Lagged effects of drought on vegetation dynamics

3.2.1. Temporal variation of time-lag effects

As indicated by the $R_{\max_lag}$, the month of lagged effects corresponding to the highest correlation coefficient pointed to vegetation sensitivity to drought. The shorter the lagged month, the stronger the sensitivity of vegetation. As illustrated in Fig. 6(a), the extent and intensity of the effects of lagged drought on vegetation varied across climatic zones. Over 30 % of vegetated zones in the arid region were significantly affected by drought, with the effects exhibiting a lag of two to three months, during which the maximum correlation coefficient was also observed. These results indicate that vegetation in the arid zone was more susceptible to short-term drought (one–three months). Contrastingly, six months lagged drought significantly affected nearly 30 % of vegetated areas in the semi-arid zone. In the semi-humid zone, vegetation was more sensitive to prolonged drought, as evidenced by the NDVI being most widely and strongly affected by drought with effects that lagged for 11 months.

Moreover, the sensitivity of vegetation to drought differed by vegetation type [Fig. 6(b)]. Alpine vegetation and meadows were the most sensitive to drought, as the highest correlation coefficients and the largest significantly correlated areas were observed between the NDVI and SPEI during the same periods. Grasslands responded most strongly to drought with a lag of two months, whereas underbrush was the most sensitive to drought with a lag of six months. The crop types dominantly cultivated in the YRB during the studied period were winter wheat and summer maize, and the optimal crop responses to drought occurred at time lags of six and three months. Shrubs, coniferous forests, and broadleaf forests were insensitive to drought, as demonstrated by the largest areas of significant correlation and correlation coefficients recorded at long-term lags (10–12 months).

3.2.2. Spatial patterns of time-lag effects

Based on the determination of the $R_{\max_lag}$ for each pixel, we obtained the time-lag spatial patterns of vegetation response to drought in the YRB (Fig. 4). Overall, 91.8 % of vegetated areas exhibited effects from lagged drought, with 60.6 % of the time-lagged responses being significant [Fig. 7(a)]. The NDVI was significantly promoted by lagged drought in 32.2 % of vegetated areas, mostly in the central arid and semi-arid zones. By contrast, the NDVI was negatively affected by the lagged SPEI in 28.4 % of the YRB, principally in the northeast area of the semi-arid zone and the southwest region of the semi-humid zone.

The lagged drought affected vegetation typically occurred within six months in the YRB (Fig. 4b). The area with no lag effects accounted for 8.2 % of the YRB and was primarily situated in the high-altitude areas of

the upper basin [Fig. 7(b)]. The time lag of the most optimal response was six months, which occurred in 20.9 % of vegetated areas in the YRB, followed by two and three months occurring in 16.3 % and 15.8 % of vegetated areas, respectively. The area with a three-month time lag of optimal response accounted for 34.6 % of the total, transpiring predominantly in the central and western arid and semi-arid zones. Vegetation in the northern semi-arid zone responded more strongly to six months lagged drought, whereas vegetation in the southern semi-humid zone responded to long-term lagged drought (10–12 months).

3.2.3. Relationship between water availability and time-lag effects

The relationship between the mean annual SPEI and mean $R_{\max_lag}$, as well as the corresponding mean months of lagged effects for each vegetation type in different climatic zones, is illustrated in Fig. 5. The mean $R_{\max_lag}$ decreased significantly with increasing annual water balance ($R^2 = 0.30$, $p < 0.01$), suggesting that vegetation in the arid region was more susceptible to drought stress [Fig. 8(a)]. A significant inverted U-shaped relationship was found between months of lagged effects and annual mean SPEI ($R^2 = 0.55$, $p < 0.001$), with the lagged months being significantly shorter as water availability improved [Fig. 8(b)]. As water balance conditions improved, the mean months of lagged effects gradually increased in the arid zone (mean of approximately-four months), peaking in coniferous and broadleaf forests in the semi-humid zone (mean of approximately 10 months) and then decreasing in alpine vegetation and meadows. Except for meadows and alpine vegetation at high altitudes, months of lagged effects were normally ranked in the order semi-humid zone > semi-arid zone > arid zone, indicating that the month of lagged effects on drought-affected vegetation lengthens with enhanced water balance.

4. Discussion

4.1. Cumulative effects of drought on vegetation

We confirmed previous findings that vegetation responds asymmetrically to drought with temporal effects (Huang et al., 2018; Wei et al., 2022a; Zhang et al., 2022). Nevertheless, our study deviated from typical scholarship in that we invested additional effort into investigating the differences in spatiotemporal patterns of lagged and accumulated drought affecting vegetation across climate zones and vegetation types (Figs. S1 and S2). Vegetation tolerance to drought, which was determined via the months of optimal response to cumulative effects under the NDVI and one- to 12-month SPEI timescale for each pixel, differed depending on climate zone and vegetation type (Li et al., 2018; Xu et al., 2018). The differences in months of accumulated effects

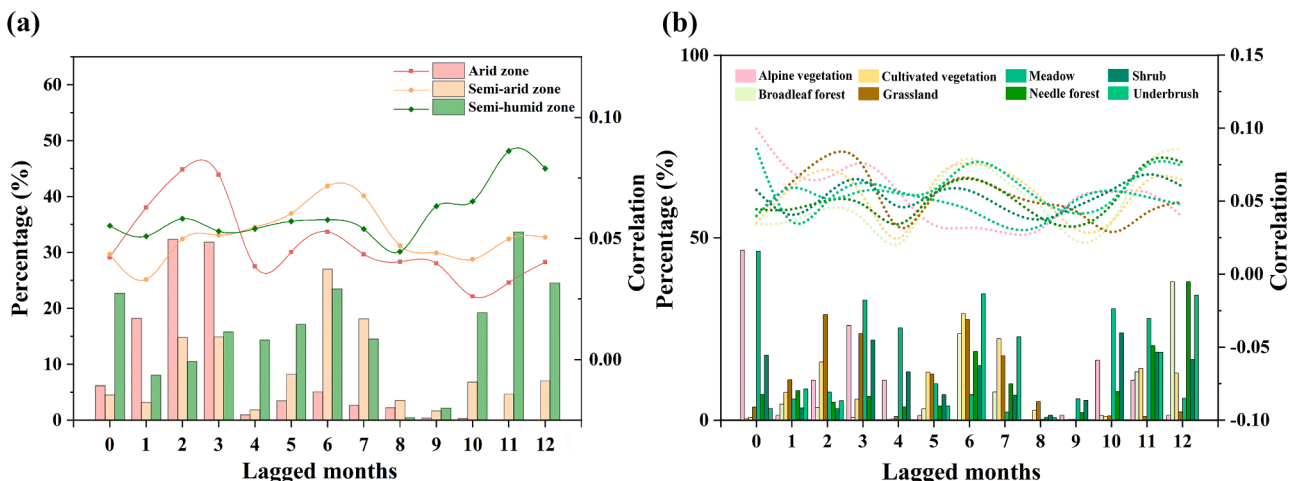


Fig. 6. Temporal variations in the (a) mean absolute values of the $R_{\max_lag}$ and (b) the areas of significant correlations between the NDVI and lagged SPEI.

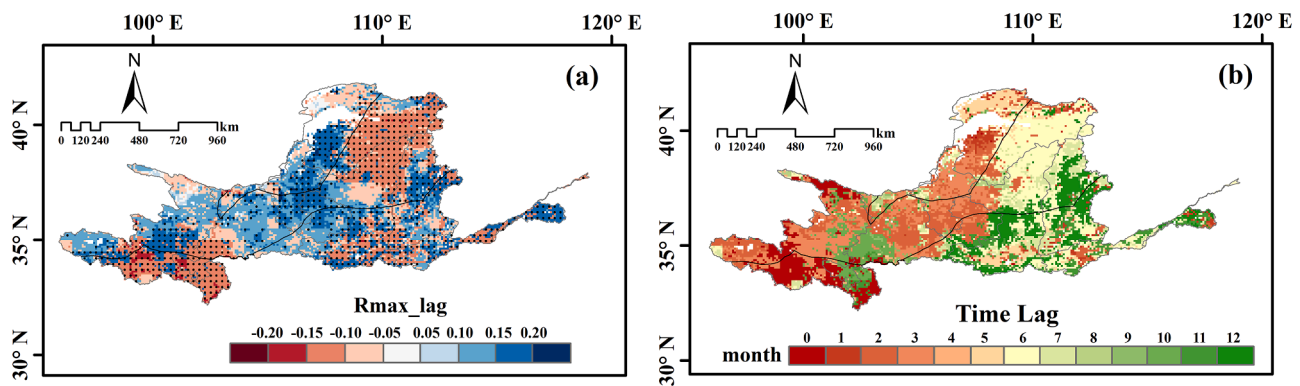


Fig. 7. Spatial patterns of (a) the $R_{\max_lag}$ and (b) corresponding lagged effects in relation to the $R_{\max_lag}$.

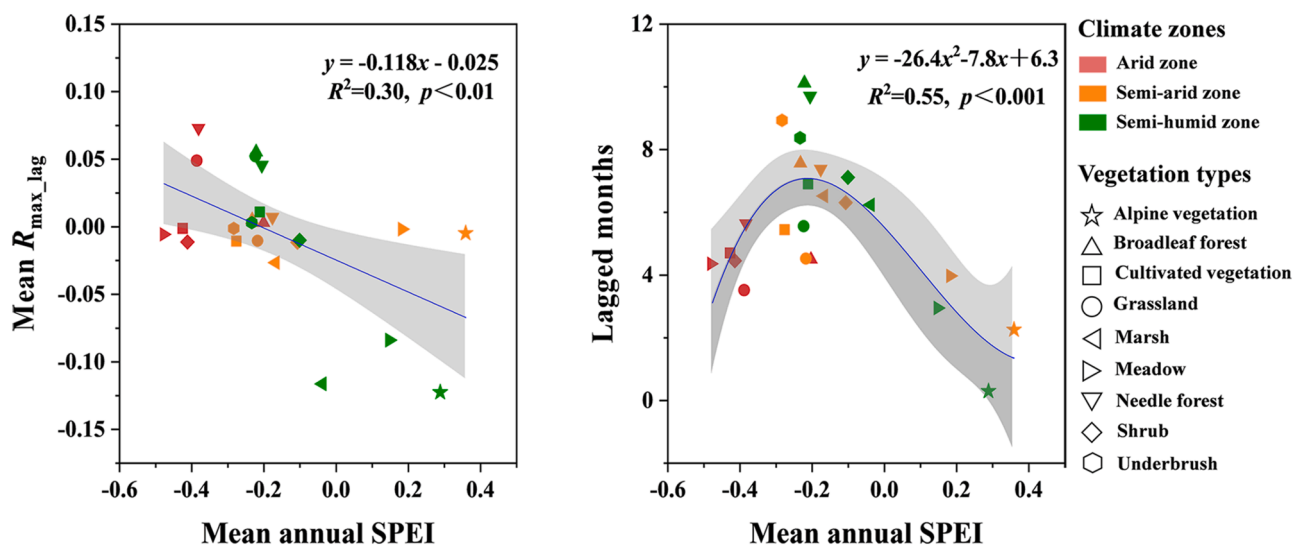


Fig. 8. The relationship between the 12-month timescale SPEI in December and (a) the mean values of $R_{\max_lag}$ and (b) the corresponding mean months of lagged effects for different vegetation types across climatic zones. The symbols represent different vegetation types, the colors indicate different climatic zones, and the gray shaded areas refer to the 95% confidence intervals of the fitted lines.

reflected the varying strategies of diverse vegetation types for coping with water deficits in diverse climatic zones (Fang et al., 2019b). In general, the accumulated effects of drought on vegetation in the arid, semi-arid, and semi-humid zones of the YRB occurred primarily at six to eight, four to six, and seven to eight months, respectively [Fig. 3(a)]. In arid zones, vegetation suffers from long-term water shortage, which is the key constraint on vegetation growth; such growth is facilitated by increased precipitation (soil moisture) over a given period (Fang et al., 2019a; Xu et al., 2018). The physiological characteristics and functional strategies of vegetation in arid zones for coping with water deficits have evolved, enabling allowing them to respond rapidly to variations in water availability; these characteristics and strategies include the reduction of water loss, respiratory depletion, and photosynthesis (Chaves et al., 2002; Peguero-Pina et al., 2020; Yang et al., 2020). These conditions differ from those in semi-humid regions, where soil moisture is relatively abundant and atmospheric moisture deficits are transmitted to soil over a long time, thereby diminishing vegetation tolerance for drought (Cheng et al., 2021; Sun et al., 2021).

Furthermore, the cumulative effects of drought on vegetation in the YRB varied across vegetation types. Overall, vegetation tolerance for drought was ranked as follows: grasslands, cultivated lands, and forests—a trend that may be relevant to root characteristics and water use patterns (Ma et al., 2018; Peng et al., 2019). As a deep-rooted species, the forests in the YRB tended to tap soil water in deeper layers to alleviate water deficits under extreme droughts (Andrade et al., 2005;

Davidson et al., 2000). Frequently, atmospheric drought initially led to a deficit in the soil water content of the surface layer, which was then gradually transferred to deeper soil layers, thereby prolonging the months of accumulated drought (Fan et al., 2017; Xu et al., 2018). Conversely, short roots enabled grasses to capture water exclusively from the surface and mid-soil layers and thus reacted quickly to atmospheric water deficits (Dodd et al., 1998; Knapp et al., 2015). Although cultivated vegetation was a shallow-rooted species, it exhibited longer months of lagged drought effects than those shown by grasslands, which may be explained by irrigation that weakens drought sensitivity (Deng et al., 2006). On this basis, there may be a connection between vegetation sensitivity to drought and anthropogenic activities, such as the breeding of drought-resistant crops and the implementation of irrigation measures (Fang et al., 2019a; Xu et al., 2018). Specifically, significant negative correlations were found between the NDVI and simultaneous SPEI in alpine vegetation and meadows, demonstrating that vegetation in these regions was highly susceptible to climate change (Ding et al., 2020; Verrall and Pickering, 2020). As these regions were characterized by high altitudes and low temperatures, excessive precipitation would have decreased temperature and solar radiation, as well as possibly caused frost damage, severely limiting vegetation growth (Ge et al., 2021; Zhan et al., 2022).

Annual hydrological conditions may have been a major contributor to the cumulative impact of drought on vegetation (Peng et al., 2019; Zhang et al., 2017). For the most of the YRB, the correlation between the

NDVI and annual SPEI weakened with increasing annual water availability and shifted from a positive to a negative correlation (Fig. 5). The arid zone (which had a low annual SPEI) experienced stronger cumulative drought effects, further confirming the vegetation tolerance to water availability in the arid zone (Liu et al., 2018). Increasing water availability promoted the physiological activities of vegetation in the arid zone, as evidenced by the increase in the NDVI (Vicente-Serrano et al., 2013). From another perspective, the months of vegetation response to accumulated drought increased and then decreased along with a rise in the annual SPEI, with the peak occurring in the semi-humid forest ecosystems. Therefore, vegetation in arid zones can rapidly respond to short-term water deficits (Dorman et al., 2013; Shi et al., 2021), while forests in semi-humid zones are more resilient to droughts because of both their surrounding environment and root systems (Vicente-Serrano et al., 2013; Wang et al., 2022; Yan et al., 2017).

4.2. Lagged effects of drought on vegetation

Although cumulative drought may have a legacy effect on vegetation, we characterized the sensitivity of vegetation to drought through correlations between NDVI and the one-month timescale SPEI. This was because to determine the sensitivity of vegetation to drought as quickly as possible, we must capture each of the fastest occurring droughts (i.e. the droughts occurring on the shortest timescale). As the SPEI described droughts based on monthly water balances (precipitation versus evapotranspiration), its shortest timescale was one month. Additionally, cumulative droughts developed from droughts that occurred on a one-month timescale, while the occurrence of a monthly drought did not necessarily indicate a cumulative drought. Consequently, a cumulative SPEI may obscure the fact that a drought occurred over one month. We therefore examined the lagged effects of drought on vegetation using a one-month timescale SPEI.

We found that lagged drought affected more than 90 % of the vegetated areas in the YRB and that the effects were temporal and spatially heterogeneous (Figs. 6 and 7). Consequently, water availability at a previous stage could have affected vegetation growth at the present (Ji and Peters, 2003; Peng et al., 2019). Altogether, the lagged effects of drought on vegetation in the arid, semi-arid, and semi-humid zones occurred primarily in the short (one–three months), medium (five–seven months), and long (10–12 months) terms, respectively [Fig. 6(a)]. In addition, grasslands, cultivated vegetation, and forests were sensitive to short- (one–three months), medium- (five–seven months), and long-term (10–12 months) lagged droughts, respectively. Although the temporal patterns of these lag effects in different climatic conditions and biomes were similar to the cumulative effects, the time-lag effects exhibited stronger temporal fluctuations (Wu et al., 2018). Perhaps in the process of transmission, the lagged effects of drought may have been affected by the sensitivity of species to drought, as well as local environmental changes (Anderson et al., 2010; Du et al., 2014; Wu et al., 2015). The cumulative effects were stronger than the lagged effects in most areas of the YRB, independent of the climate zone and vegetation type. This result implies that cumulative water availability in a given period had a greater impact on vegetation growth than availability at individual time points (Wei et al., 2022b).

A lagged effect of drought on vegetation was similar to a cumulative effect under hydrological conditions. There was a decrease in the intensity of NDVI responses to annual SPEI changes as the climate became wetter, indicating that vegetation in the arid zone was more sensitive to droughts (Vicente-Serrano et al., 2013). Additionally, the NDVI tended to be affected by drought in longer months of accumulated events in the humid zone because short-term drought did not readily produce changes in the NDVI under a long-term water surplus (Fang et al., 2019a).

4.3. Strengths and limitations

Although our study has provided insight into the differential

responses of various climatic zones and ecosystems to drought and their drought tolerance, it is encumbered with some limitations. Firstly, the correlation analysis was conducted to quantify the drought-vegetation relationships without accounting for the nonlinear relationship between them. Then, to focus on the dominant patterns of drought-vegetation relationships, we have retained only the strongest patterns of influence and ignored other degree-based influences, which may result in the loss of information about other impacts. Thus, future studies may use principal component analysis, for example, to identify more patterns of vegetation response to drought and to retain as much information as possible regarding vegetation-drought interactions. Additionally, we focused only on the effects of drought on vegetation rather than other climatic extremes (e.g., flooding), human activities (e.g., irrigation), or natural disturbances (e.g., pests and diseases). Further research on the influence of climatic and non-climatic factors on vegetation change using machine learning may be worthwhile.

5. Conclusion

This study analyzed the spatiotemporal patterns of drought-induced cumulative and time-lag effects on vegetation across different climatic zones and vegetation types in the YRB using the monthly NDVI, one-month SPEI timescale (reflecting lagged drought), and one- to 12-month SPEI timescale (reflecting accumulated drought). According to the maximum Pearson correlation coefficients between current monthly NDVI and accumulated (lagged) SPEI and their corresponding months of accumulated (lagged) effects, we shed light on the sensitivity to and tolerance for drought of vegetation in the region of interest. We found that vegetation growth in most of the YRB (77.4 %) depended on water resources, with the effects on 40.4 % of vegetated areas significantly facilitated by the increase in accumulated SPEI. The longest duration of cumulative effects was identified in the semi-humid zone (eight months), followed by the arid zone (seven months) and semi-arid zone (five months). In terms of main vegetation types, forests exhibited the strongest drought tolerance, followed by cultivated croplands and grasslands, whereas alpine vegetation showed the weakest drought tolerance. Vegetated areas in 91.8 % of the YRB were affected by lagged drought, with 60.6 % of the effects being significant. The arid, semi-arid, and semi-humid zones were more sensitive to short- (one–three months), medium- (five–seven months), and long-term (10–12 months) lagged drought, respectively, and grasslands were more sensitive to drought than were cultivated vegetation and forests. Annual water availability in the YRB significantly influenced the relationship between vegetation and drought, with the sensitivity to and tolerance of vegetation for accumulated (lagged) drought weakening with increasing water availability. In general, the intensity of the cumulative effects and the corresponding months of optimal response was stronger than that of the lagged effects, regardless of climatic condition and vegetation type.

CRediT authorship contribution statement

Cun Zhan: Visualization, Writing – original draft. **Chuan Liang:** Writing – review & editing, Supervision. **Lu Zhao:** Writing – review & editing, Funding acquisition. **Shouzheng Jiang:** Conceptualization, Methodology. **Kaijie Niu:** Methodology, Validation. **Yaling Zhang:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

A special thanks are due to the GIMMS group for supplying the GIMMS NDVI 3 g.v1 dataset. The Spanish National Research Council (CSIC) is sincerely thanked for sharing the SPEI dataset with us. We sincerely appreciate the support of the Resource and Environment Science and Data Center to the vegetation type dataset. Funding for this work has been provided by the National Natural Science Foundation of China (5210090533).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2022.109409>.

References

- AghaKouchak, A., Chiang, F., Huning, L.S., Love, C.A., Mallakpour, I., Mazdiyasn, O., Moftakhari, H., Papalexio, S.M., Ragno, E., Sadegh, M., 2020. Climate extremes and compound hazards in a warming world. *Annu. Rev. Earth Planet. Sci.* 48, 519–548.
- Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. *Crop evapotranspiration-Guidelines for computing crop water requirements-FAO Irrigation and drainage paper 56*. Fao, Rome 300, D05109.
- Anderegg, W.R., Trugman, A.T., Badgley, G., Konings, A.G., Shaw, J., 2020. Divergent forest sensitivity to repeated extreme droughts. *Nat. Clim. Change* 10, 1091–1095.
- Anderson, L.O., Malhi, Y., Aragão, L.E., Ladle, R., Arai, E., Barbier, N., Phillips, O., 2010. Remote sensing detection of droughts in Amazonian forest canopies. *New Phytol.* 187, 733–750.
- Andrade, J.L., Meinzer, F.C., Goldstein, G., Schnitzer, S.A., 2005. Water uptake and transport in lianas and co-occurring trees of a seasonally dry tropical forest. *Trees* 19, 282–289.
- Bellprat, O., Guemas, V., Doblas-Reyes, F., Donat, M.G., 2019. Towards reliable extreme weather and climate event attribution. *Nat. Commun.* 10, 1–7.
- Chaves, M.M., Pereira, J.S., Maroco, J., Rodrigues, M.L., Ricardo, C.P., Osório, M.L., Carvalho, I., Faria, T., Pinheiro, C., 2002. How plants cope with water stress in the field. In: *Photosynthesis and Growth*. *Ann Bot* 89 Spec No, pp. 907–916.
- Cheng, Z., Cui, Z., Shi, J., Liu, Y., La Pierre, K.J., Wu, G.-L., 2021. Plant functional types drive differential responses of grassland ecosystem functions along a precipitation gradient. *Ecol. Ind.* 133, 108433.
- Davidson, E.A., Verchot, L.V., Cattaneo, J.H., Ackerman, I.L., Carvalho, J., 2000. Effects of soil water content on soil respiration in forests and cattle pastures of eastern Amazonia. *Biogeochemistry* 48, 53–69.
- Deng, X.-P., Shan, L., Zhang, H., Turner, N.C., 2006. Improving agricultural water use efficiency in arid and semiarid areas of China. *Agric. Water Manag.* 80, 23–40.
- Ding, Y., Li, Z., Peng, S., 2020. Global analysis of time-lag and-accumulation effects of climate on vegetation growth. *Int. J. Appl. Earth Obs. Geoinf.* 92, 102179.
- Dodd, M., Lauenroth, W., Welker, J., 1998. Differential water resource use by herbaceous and woody plant life-forms in a shortgrass steppe community. *Oecologia* 117, 504–512.
- D'Orangeville, L., Maxwell, J., Kneeshaw, D., Pederson, N., Duchesne, L., Logan, T., Houle, D., Arseneault, D., Beier, C.M., Bishop, D.A., 2018. Drought timing and local climate determine the sensitivity of eastern temperate forests to drought. *Glob. Change Biol.* 24, 2339–2351.
- Dorman, M., Svoray, T., Perevolotsky, A., Sarris, D., 2013. Forest performance during two consecutive drought periods: diverging long-term trends and short-term responses along a climatic gradient. *For. Ecol. Manage.* 310, 1–9.
- Du, J., He, Z., Yang, J., Chen, L., Zhu, X., 2014. Detecting the effects of climate change on canopy phenology in coniferous forests in semi-arid mountain regions of China. *Int. J. Remote Sens.* 35, 6490–6507.
- Fan, Y., Miguez-Macho, G., Jobbágy, E.G., Jackson, R.B., Otero-Casal, C., 2017. Hydrologic regulation of plant rooting depth. *Proc. Natl. Acad. Sci.* 114, 10572–10577.
- Fang, W., Huang, S., Huang, Q., Huang, G., Wang, H., Leng, G., Wang, L., Guo, Y., 2019a. Probabilistic assessment of remote sensing-based terrestrial vegetation vulnerability to drought stress of the Loess Plateau in China. *Remote Sens. Environ.* 232, 111290.
- Fang, W., Huang, S., Huang, Q., Huang, G., Wang, H., Leng, G., Wang, L., Li, P., Ma, L., 2019b. Bivariate probabilistic quantification of drought impacts on terrestrial vegetation dynamics in mainland China. *J. Hydrol.* 577, 123980.
- Fang, Y., Xiong, L., 2015. General mechanisms of drought response and their application in drought resistance improvement in plants. *Cell. Mol. Life Sci.* 72, 673–689.
- Gao, Y., Jia, J., Lu, Y., Yang, T., Lyu, S., Shi, K., Zhou, F., Yu, G., 2021. Determining dominating control mechanisms of inland water carbon cycling processes and associated gross primary productivity on regional and global scales. *Earth Sci. Rev.* 213, 103497.
- Ge, W., Deng, L., Wang, F., Han, J., 2021. Quantifying the contributions of human activities and climate change to vegetation net primary productivity dynamics in China from 2001 to 2016. *Sci. Total Environ.* 773, 145648.
- Geng, G., Yang, R., Liu, L., 2022. Downscaled solar-induced chlorophyll fluorescence has great potential for monitoring the response of vegetation to drought in the Yellow River Basin, China: Insights from an extreme event. *Ecol. Ind.* 138, 108801.
- Hua, L., Wang, H., Sui, H., Wardlow, B., Hayes, M.J., Wang, J., 2019. Mapping the spatial-temporal dynamics of vegetation response lag to drought in a semi-arid region. *Remote Sens.* 11, 1873.
- Huang, M., Wang, X., Keenan, T.F., Piao, S., 2018. Drought timing influences the legacy of tree growth recovery. *Glob. Change Biol.* 24, 3546–3559.
- Ji, H., Chen, S., Pan, S., Xu, C., Jiang, C., Fan, Y., 2018. Morphological variability of the active Yellow River mouth under the new regime of riverine delivery. *J. Hydrol.* 564, 329–341.
- Ji, L., Peters, A.J., 2003. Assessing vegetation response to drought in the northern Great Plains using vegetation and drought indices. *Remote Sens. Environ.* 87, 85–98.
- Jiang, P., Ding, W., Yuan, Y., Ye, W., 2020. Diverse response of vegetation growth to multi-time-scale drought under different soil textures in China's pastoral areas. *J. Environ. Manage.* 274, 110992.
- Jiang, W., Niu, Z., Wang, L., Yao, R., Gui, X., Xiang, F., Ji, Y., 2022. Impacts of drought and climatic factors on vegetation dynamics in the Yellow River Basin and Yangtze River Basin, China. *Remote Sens.* 14, 930.
- Jiang, Y., Wang, R., Peng, Q., Wu, X., Ning, H., Li, C., 2018. The relationship between drought activity and vegetation cover in Northwest China from 1982 to 2013. *Nat. Hazards* 92, 145–163.
- Jia, L., Yu, K., Li, Z., Li, P., Xu, G., Cheng, Y., Zhang, X., Yang, Z., 2021. The effect of meteorological drought on vegetation cover in the Yellow River basin, China. *Int. J. Climatol.*
- Jin, K., Wang, F., Zong, Q., Qin, P., Liu, C., Wang, S., 2022. Spatiotemporal differences in climate change impacts on vegetation cover in China from 1982 to 2015. *Environ. Sci. Pollut. Res.* 29, 10263–10276.
- Knapp, A.K., Carroll, C.J., Denton, E.M., La Pierre, K.J., Collins, S.L., Smith, M.D., 2015. Differential sensitivity to regional-scale drought in six central US grasslands. *Oecologia* 177, 949–957.
- Li, C., Leal Filho, W., Yin, J., Hu, R., Wang, J., Yang, C., Yin, S., Bao, Y., Ayal, D.Y., 2018. Assessing vegetation response to multi-time-scale drought across inner Mongolia plateau. *J. Cleaner Prod.* 179, 210–216.
- Liu, Y., Li, Z., Chen, Y., 2021. Continuous warming shift greening towards browning in the Southeast and Northwest High Mountain Asia. *Sci. Rep.* 11, 1–14.
- Liu, L., Zhang, Y., Wu, S., Li, S., Qin, D., 2018. Water memory effects and their impacts on global vegetation productivity and resilience. *Sci. Rep.* 8, 1–9.
- Luo, M., Sa, C., Meng, F., Duan, Y., Liu, T., Bao, Y., 2020. Assessing extreme climatic changes on a monthly scale and their implications for vegetation in Central Asia. *J. Cleaner Prod.* 271, 122396.
- Ma, Z., Guo, D., Xu, X., Lu, M., Bardgett, R.D., Eissenstat, D.M., McCormack, M.L., Hedin, L.O., 2018. Evolutionary history resolves global organization of root functional traits. *Nature* 555, 94–97.
- Mohammad, A., Wang, X., Xu, X., Peng, L., Yang, Y., Zhang, X., Myneni, R.B., Piao, S., 2013. Drought and spring cooling induced recent decrease in vegetation growth in Inner Asia. *Agric. For. Meteorol.* 178, 21–30.
- Peguero-Pina, J.J., Vilagrosa, A., Alonso-Forn, D., Ferrio, J.P., Sancho-Knapik, D., Gil-Pelegrín, E., 2020. Living in drylands: Functional adaptations of trees and shrubs to cope with high temperatures and water scarcity. *Forests* 11, 1028.
- Peng, J., Wu, C., Zhang, X., Wang, X., Gonsamo, A., 2019. Satellite detection of cumulative and lagged effects of drought on autumn leaf senescence over the Northern Hemisphere. *Glob. Change Biol.* 25, 2174–2188.
- Qi, G., Song, J., Li, Q., Bai, H., Sun, H., Zhang, S., Cheng, D., 2022. Response of vegetation to multi-timescales drought in the Qinling Mountains of China. *Ecol. Ind.* 135, 108539.
- Qin, G., Meng, Z., Fu, Y., 2022. Drought and water-use efficiency are dominant environmental factors affecting greenness in the Yellow River Basin, China. *Sci. Total Environ.* 834, 155479.
- Shi, S., Yu, J., Wang, F., Wang, P., Zhang, Y., Jin, K., 2021. Quantitative contributions of climate change and human activities to vegetation changes over multiple time scales on the Loess Plateau. *Sci. Total Environ.* 755, 142419.
- Sun, S., Du, W., Song, Z., Zhang, D., Wu, X., Chen, B., Wu, Y., 2021. Response of gross primary productivity to drought time-scales across China. *J. Geophys. Res.: Biogeosci.* 126, e2020JG005953.
- Ummenhofer, C.C., Meehl, G.A., 2017. Extreme weather and climate events with ecological relevance: a review. *Philos. Trans. Royal Soc. B: Biol. Sci.* 372, 20160135.
- Verrall, B., Pickering, C.M., 2020. Alpine vegetation in the context of climate change: a global review of past research and future directions. *Sci. Total Environ.* 748, 141344.
- Vicente-Serrano, S.M., Beguería, S., López-Moreno, J.I., 2010. A multiscale drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *J. Clim.* 23, 1696–1718.
- Vicente-Serrano, S.M., Gouveia, C., Camarero, J.J., Beguería, S., Trigo, R., López-Moreno, J.I., Azorín-Molina, C., Pasho, E., Lorenzo-Lacruz, J., Revuelto, J., 2013. Response of vegetation to drought time-scales across global land biomes. *Proc. Natl. Acad. Sci.* 110, 52–57.
- Wang, H., Chen, A., Wang, Q., He, B., 2015. Drought dynamics and impacts on vegetation in China from 1982 to 2011. *Ecol. Eng.* 75, 303–307.
- Wang, B., Chen, T., Li, C., Xu, G., Wu, G., Liu, G., 2022. Discrepancy in growth resilience to drought among different stand-aged forests declines going from a semi-humid region to an arid region. *For. Ecol. Manage.* 511, 120135.
- Wang, H., Li, Z., Cao, L., Feng, R., Pan, Y., 2021. Response of NDVI of natural vegetation to climate changes and drought in China. *Land* 10, 966.
- Wang, X., Qiu, B., Li, W., Zhang, Q., 2019. Impacts of drought and heatwave on the terrestrial ecosystem in China as revealed by satellite solar-induced chlorophyll fluorescence. *Sci. Total Environ.* 693, 133627.
- Wei, X., He, W., Zhou, Y., Ju, W., Xiao, J., Li, X., Liu, Y., Xu, S., Bi, W., Zhang, X., 2022a. Global assessment of lagged and cumulative effects of drought on grassland gross primary production. *Ecol. Ind.* 136, 108646.

- Wei, X., He, W., Zhou, Y., Ju, W., Xiao, J., Li, X., Liu, Y., Xu, S., Bi, W., Zhang, X., Cheng, N., 2022b. Global assessment of lagged and cumulative effects of drought on grassland gross primary production. *Ecol. Ind.* 136, 108646.
- Wu, X., Liu, H., Li, X., Ciais, P., Babst, F., Guo, W., Zhang, C., Magliulo, V., Pavelka, M., Liu, S., 2018. Differentiating drought legacy effects on vegetation growth over the temperate Northern Hemisphere. *Glob. Change Biol.* 24, 504–516.
- Wu, D., Zhao, X., Liang, S., Zhou, T., Huang, K., Tang, B., Zhao, W., 2015. Time-lag effects of global vegetation responses to climate change. *Glob. Change Biol.* 21, 3520–3531.
- Xu, C., McDowell, N.G., Fisher, R.A., Wei, L., Sevanto, S., Christoffersen, B.O., Weng, E., Middleton, R.S., 2019. Increasing impacts of extreme droughts on vegetation productivity under climate change. *Nat. Clim. Change* 9, 948–953.
- Xu, H.-J., Wang, X.-P., Zhao, C.-Y., Yang, X.-M., 2018. Diverse responses of vegetation growth to meteorological drought across climate zones and land biomes in northern China from 1981 to 2014. *Agric. For. Meteorol.* 262, 1–13.
- Xu, H.-J., Wang, X.-P., Zhao, C.-Y., 2021. Drought sensitivity of vegetation photosynthesis along the aridity gradient in northern China. *Int. J. Appl. Earth Obs. Geoinf.* 102, 102418.
- Xueyu, H., 2019. 1:1 million vegetation map of China, in: National Tibetan Plateau Data, C. (Ed.). National Tibetan Plateau Data Center.
- Yan, W., Zhong, Y., Shangguan, Z., 2017. Responses of different physiological parameter thresholds to soil water availability in four plant species during prolonged drought. *Agric. For. Meteorol.* 247, 311–319.
- Yang, M.-H., Jahufer, M.Z., He, J., Dong, R., Hofmann, R., Siddique, K.H., Li, F.-M., 2020. Effect of traditional soybean breeding on water use strategy in arid and semi-arid areas. *Eur. J. Agron.* 120, 126128.
- Ying, H., Zhang, H., Zhao, J., Shan, Y., Zhang, Z., Guo, X., Rihan, W., Deng, G., 2020. Effects of spring and summer extreme climate events on the autumn phenology of different vegetation types of Inner Mongolia, China, from 1982 to 2015. *Ecol. Ind.* 111, 105974.
- Zhan, C., Liang, C., Zhao, L., Jiang, S., Niu, K., Zhang, Y., Cheng, L., 2022. Vegetation dynamics and its response to climate change in the Yellow River Basin, China. *Front. Environ. Sci.* 450.
- Zhang, Z., Ju, W., Zhou, Y., Li, X., 2022. Revisiting the cumulative effects of drought on global gross primary productivity based on new long-term series data (1982–2018). *Glob. Change Biol.* 28, 3620–3635.
- Zhang, Q., Kong, D., Singh, V.P., Shi, P., 2017. Response of vegetation to different time-scales drought across China: spatiotemporal patterns, causes and implications. *Global Planet. Change* 152, 1–11.
- Zhao, J., Feng, H., Xu, T., Xiao, J., Guerrieri, R., Liu, S., Wu, X., He, X., He, X., 2021. Physiological and environmental control on ecosystem water use efficiency in response to drought across the northern hemisphere. *Sci. Total Environ.* 758, 143599.
- Zhao, A., Zhang, A., Cao, S., Liu, X., Liu, J., Cheng, D., 2018. Responses of vegetation productivity to multi-scale drought in Loess Plateau, China. *Catena* 163, 165–171.
- Zhao, A., Zhang, A., Liu, J., Feng, L., Zhao, Y., 2019. Assessing the effects of drought and “Grain for Green” Program on vegetation dynamics in China’s Loess Plateau from 2000 to 2014. *Catena* 175, 446–455.
- Zhao, A., Yu, Q., Feng, L., Zhang, A., Pei, T., 2020. Evaluating the cumulative and time-lag effects of drought on grassland vegetation: a case study in the Chinese Loess Plateau. *J. Environ. Manage.* 261, 110214.
- Zheng, D., 1999. A study on the eco-geographic regional system of China, FAO FRA2000 Global Ecological Zoning Workshop, Cambridge, UK.
- Zuo, D., Han, Y., Xu, Z., Li, P., Ban, C., Sun, W., Pang, B., Peng, D., Kan, G., Zhang, R., 2021. Time-lag effects of climatic change and drought on vegetation dynamics in an alpine river basin of the Tibet Plateau, China. *J. Hydrol.* 600, 126532.